Detecting model misspecification in field-level cosmic-shear inference: cross-encoder posterior disagreement versus summary-space out-of-distribution scores

Abstract

Neural posterior estimation freezes the forward model into the network, so an observation containing physics absent from the training simulations yields a confidently wrong posterior with no internal warning. We compare two label-free misspecification detectors on a production KiDS-Legacy-like field-level model trained on Gower Street N-body mocks: the disagreement (a Kullback–Leibler statistic) between the posteriors of five independently seeded encoders, and an out-of-distribution (OOD) score of the observation's learned 8-dimensional summary against the training summary cloud. Six paired test suites probe the intrinsic-alignment (IA) model, survey depth and the source galaxy bias b_g. On the IA suites the single-encoder kNN OOD score wins (AUROC 0.93 vs. 0.89 for the five-encoder KL; power 0.80 vs. 0.74 at 5% false alarms) at a fifth of the training cost, and needs only a few hundred reference summaries. An IA amplitude offset of $|\Delta A_{\rm IA}^{\rm total}| \simeq 0.8$–$1.0$ from the training median is detected in half of all mocks, $\simeq 1.1$–1.3 in 80%; a change of IA model form at matched amplitude is invisible. Depth and b_g variations are not detected. The b_g suites are at most mildly outside the training prior and their calibration is at the in-distribution floor, so the implicit marginalisation over b_g has not failed; yet the posterior mean tracks the true b_g, shifting by $0.8\,\sigma_{\rm post}$ in $\Omega_{\rm m}$ and σ_8 in opposite directions for $b_g = 1.0$ and 1.3 ($\Delta\Omega_{\rm m} = -0.0489$ between them, seven times the DES Y3 bound). This bias, of a b_g-marginalised posterior at fixed b_g and of a coherent across-bin b_g configuration, is invisible by construction to every detector and sets the real limit of what these diagnostics protect against.

1 Motivation

Simulation-based inference with neural compression [1, 2] is now the standard route to field-level cosmology from weak-lensing maps [3]. Its price is that every simulator assumption is baked into the estimator: if the sky contains physics the training suite did not, the posterior stays narrow and its calibration, established on held-out simulations, no longer applies. A detector that flags such an event on the real observation, without knowing the true parameters, is therefore a prerequisite for a credible result. Two ideas are natural. Independently trained networks agree where the data constrain them and diverge where they extrapolate, so the disagreement of an ensemble of posteriors is a misspecification statistic [4]. Alternatively, the compressed summary of a misspecified observation should fall outside the cloud of training summaries, a classical OOD problem [5, 6]. This report puts the two head-to-head on the same suites with the same statistic, and quantifies the smallest misspecification either can see.

2 Model, test suites and detectors

Inference model. A hybrid bandpower-plus-map encoder compresses a six-bin tomographic KiDS-Legacy-like shear observation into an 8-dimensional summary t; a normalising flow estimates $p(\theta\,|\,t)$ over the seven Gower Street cosmological parameters and the two NLA-M intrinsic-alignment parameters $(A_{\rm IA}, \beta_{\rm IA})$ of [7]. Encoder and flow are pre-trained on a large log-normal suite and fine-tuned on 300 N-body cosmologies (multifidelity NPE with the VMIM compression loss). Five copies were trained from independent seeds and splits; each is a complete posterior estimator. The training simulations draw the per-bin source galaxy biases $b_g^{(i)}$ from a Gaussian prior (means 1.018–1.480, widths 0.180–0.099 from bin 1 to 6, truncated at $\pm3\sigma$), but b_g is *not* an inference parameter: the posterior is implicitly marginalised over the b_g prior.

Test suites. All suites re-simulate the same locked set of 199 held-out cosmologies with $\sim$8 footprint-rotation and shape-noise augmentations each (1590 mocks), so every comparison is paired at the cosmology level. *In-distribution*: the training physics. *NLA* and *NLA-z*: the mass-dependent NLA-M model replaced by the plain non-linear alignment model, without and with a redshift power law, at amplitudes $A_{\rm IA}^{\rm total} \in [-6, 6]$. *Variable depth*: position-dependent survey depth. $b_g = 1.0$ and $b_g = 1.3$: the galaxy bias pinned to one value in all six bins (40 cosmologies $\times$ 2 mocks). Amplitudes are compared in the KiDS-Legacy $A_{\rm IA}^{\rm total}$ convention: the NLA suites store it directly, and the NLA-M training amplitude is converted per mock, $A_{\rm IA}^{\rm total} = \langle A_{\rm IA} f_{\rm red}^{(i)} (\bar{M}_h^{(i)}/10^{13.5})^{\beta_{\rm IA}} \rangle_i$, giving a training support $A_{\rm IA}^{\rm total} \in [0.31, 0.60]$ (5–95%), median 0.43.

Detector 1: cross-encoder KL. The five posteriors of one observation are approximated by diagonal Gaussians in the scaled parameter space and scored by their mean pairwise symmetric KL divergence. It needs no reference data and no true parameters, but $K \geq 2$ trained encoders. In-distribution its mean is 0.22 nats (median 0.19, 95th percentile 0.47).

Detector 2: summary-space OOD. The summary t is whitened with the training-cloud covariance and scored by its mean distance to the $k = 10$ nearest training summaries (kNN, [5]; 19167 summaries from 240 cosmologies). A *conditional* variant first removes the parameter dependence, $r = t - \mathbb{E}[t\,|\,\theta]$, and scores the residual; it needs θ (the truth in simulations, posterior draws on real data). Scores become right-tail empirical p-values against the held-out in-distribution null. Summaries of different encoders are not commensurable, so the only cross-encoder combination is of p-values; the recommended combiner is the mean of the five calibrated p-values, recalibrated against the null ("mean-p"). Fisher, Stouffer, GLS-weighted Stouffer [8] and density-of-states [6] combiners were also tested and none beats it by more than one bootstrap standard deviation. The ceiling is structural: the five encoders' scores of the same mock are correlated at $\rho = 0.72$ (the scatter is the mock's own shape noise and cosmic variance, seen by every encoder), so any five-view combination gains at most $\sqrt{5/(1+4\rho)} = 1.14$ in signal-to-noise.

Common footing. Every detector is scored by the area under the ROC curve (AUROC, [9]) of variate against in-distribution mocks, and by its detection power at a 5% false-alarm rate (the fraction of variate mocks above the null's 95th percentile). Rows are correlated augmentations of far fewer cosmologies, so all intervals are a paired cluster bootstrap over cosmologies [10]:

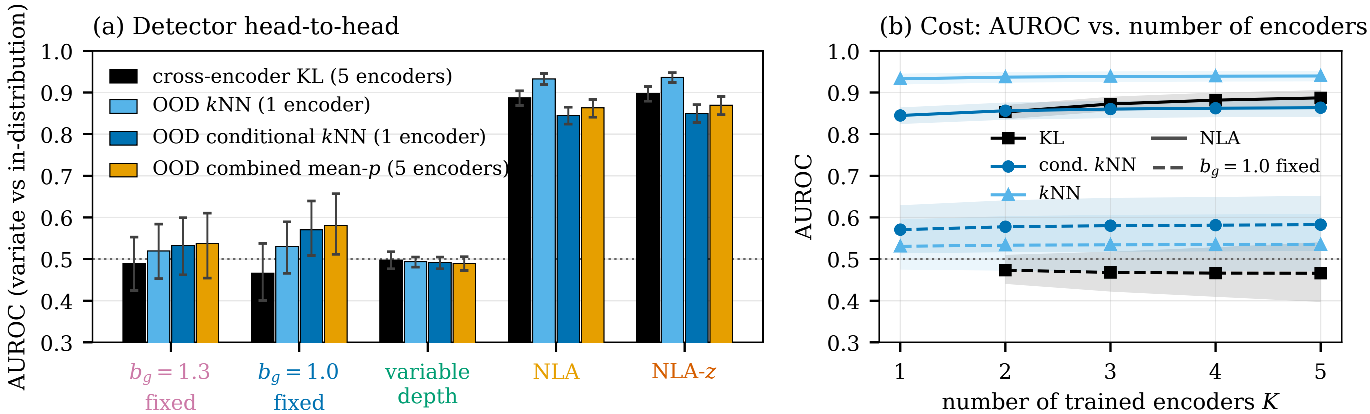

Figure 1: Head-to-head comparison and cost. (a) AUROC of each detector for separating each test suite from the in-distribution mocks (0.5 = no separation; whiskers are 95% cluster bootstrap intervals over cosmologies). Black: cross-encoder KL from five encoders; blues: single-encoder kNN and conditional-kNN OOD scores (mean over the five encoders, each scored alone); orange: the five encoders' calibrated conditional-kNN p-values combined. (b) AUROC as a function of the number of trained encoders K (KL is defined for $K \geq 2$; the OOD scores combine K encoders' p-values), for the NLA suite (solid) and the $b_g = 1.0$ suite (dashed); bands are the same bootstrap intervals.

Table 1: Detection statistics per test suite. N_{cos}: held-out cosmologies; N: mocks; $\overline{\text{KL}}$: mean cross-encoder KL (nats); $\mathcal{C}$: TARP calibration error, mean $\pm$ sd over the five encoders (in-distribution 0.0031 ± 0.0003 at $N = 1590$; floor at the b_g suites' size 0.0560 ± 0.0105). AUROC: point estimate with 95% cluster-bootstrap interval; kNN and conditional kNN (c-kNN) are single-encoder scores averaged over the five encoders; mean-p combines the five. Power: fraction of variate mocks detected at a 5% false-alarm rate. All detectors are scored on the mocks with finite posteriors in all five encoders (1558 of 1592 for NLA, 1562 for NLA-z; the other suites are complete).

					AUROC vs. in-distribution (95% CI)				power	
suite	N_{cos}	N	$\overline{\text{KL}}$	$\mathcal{C}$	KL (5 enc.)	kNN (1 enc.)	c-kNN (1 enc.)	mean-p (5 enc.)	KL	kNN
$b_g = 1.3$ fixed	40	80	0.21	0.0511 ± 0.0077	0.49 (0.42–0.55)	0.52 (0.45–0.58)	0.53 (0.46–0.60)	0.54 (0.45–0.61)	0.04	0.08
$b_g = 1.0$ fixed	40	80	0.21	0.0618 ± 0.0056	0.47 (0.40–0.54)	0.53 (0.47–0.59)	0.57 (0.51–0.64)	0.58 (0.51–0.66)	0.05	0.05
variable depth	199	1588	0.22	0.0029 ± 0.0004	0.50 (0.48–0.52)	0.49 (0.48–0.50)	0.49 (0.48–0.50)	0.49 (0.47–0.51)	0.06	0.05
NLA	199	1592	6.53	1.053 ± 0.417	0.89 (0.87–0.90)	0.93 (0.92–0.95)	0.84 (0.82–0.86)	0.86 (0.84–0.88)	0.74	0.80
NLA-z	199	1592	6.20	0.921 ± 0.434	0.90 (0.88–0.91)	0.94 (0.92–0.95)	0.85 (0.83–0.87)	0.87 (0.85–0.89)	0.75	0.82

1000 draws resample the 199 cosmologies and take all their in-distribution and variate rows (row-level intervals would be $\sim 3\times$ too narrow). As a truth-requiring, suite-level reference we also quote the TARP [11] calibration error $\mathcal{C}$.

3 Results

3.1 KL versus OOD, head-to-head

Figure 1a and Table 1 give the answer. On the two IA suites every detector separates the variate from the training physics, and the ranking is unambiguous: single-encoder kNN (0.93 [0.92, 0.95] on NLA, 0.94 on NLA-z) beats the five-encoder KL (0.89 [0.87, 0.90]; 0.90), which beats the five-encoder mean-p (0.86; 0.87) and the single-encoder conditional kNN (0.84; 0.85); the intervals of the first three do not overlap. At a fixed 5% false-alarm rate the gap widens: kNN detects 0.80 of NLA mocks, KL 0.74, mean-p 0.52, conditional kNN 0.49. The KL is heavy-tailed (mean 6.53 nats, median 2.45): many mocks sit far above the null while the rest are indistinguishable from it, and Sec. 3.2 shows the latter are those whose amplitude lies inside the training support. The detectors flag the same mocks: the Spearman correlation between per-mock KL and combined OOD p is 0.80 on NLA, although on the null the two are uncorrelated ($\rho = 0.066$ on the probit scale). Combining them (equal-weight Stouffer) does not help, AUROC 0.928 against 0.933 for kNN alone: both respond to the same physical departure and the KL adds noise, not information.

Conditional kNN ranks last on the IA suites for a definitional reason: the NLA amplitude is a different parameter from NLA-M's, cannot be conditioned on, and its spread is absorbed into the conditional null. It is, however, the only score with a hint of signal on $b_g = 1.0$ (AUROC 0.57 [0.51, 0.64]; mean-p 0.58 [0.51, 0.66]), with the lower bound just above 0.5. Every other interval on $b_g = 1.0$, $b_g = 1.3$ and variable depth contains 0.5, and the detection powers equal the false-alarm rate: clean null results (Sec. 3.3).

Cost. The KL needs at least two encoders and improves slowly with more (Fig. 1b: 0.85 at $K = 2$, 0.87 at $K = 3$, 0.89 at $K = 5$). One encoder's kNN score already beats five encoders' KL, and pooling five encoders' kNN p-values adds only +0.007 (0.940), the 1.14× ceiling above. The reference cloud can be tiny (Fig. 3): refitting the whitening and tree on random subsets of the training summaries, the NLA AUROC is 0.875 with 5 training cosmologies (400 summaries), 0.918 with 20, and saturates at 0.928 by 40; 100 random summaries spanning all cosmologies give 0.923 against 0.934 for the full cloud, and the verdict on the hard suites is unchanged at every size. One encoder plus a few hundred reference summaries is therefore both the cheaper and the stronger detector for this class of misspecification.

3.2 How much misspecification is detectable

The IA suites span $A_{\text{IA}}^{\text{total}} \in [-6, 6]$ continuously, which makes the amplitude a severity axis. Figure 2a shows that the KL is not a function of "NLA versus NLA-M" but of *how far the amplitude lies from the training support*: its running median is lowest at $A_{\text{IA}}^{\text{total}} = 0.35$ (NLA) and 0.40 (NLA-z), next to the training median 0.43, where it is 0.17 nats, below the in-distribution median. Inside the training band the detection power is 0.10 [0.01, 0.21] for the KL and 0.08 [0.01, 0.17] for kNN (50 NLA mocks),

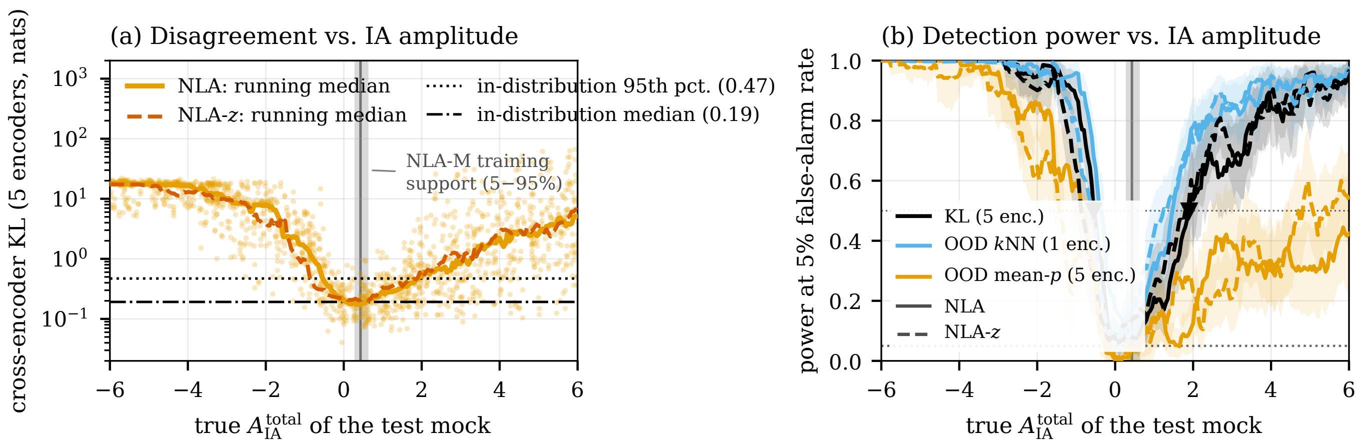

Figure 2: Detectability as a function of the intrinsic-alignment amplitude. (a) Per-mock cross-encoder KL of the NLA suite against the mock's true $A_{\rm IA}^{\rm total}$ (points), with running medians (window half-width 0.30) for NLA (solid) and NLA-z (dashed). The grey band is the 5–95% support of the NLA-M training suite converted to the same convention, the vertical line its median; horizontal lines mark the median and 95th percentile of the KL on in-distribution mocks. (b) Detection power at a 5% false-alarm rate in the same sliding window for the five-encoder KL (black), the single-encoder kNN score (light blue, averaged over encoders) and the five-encoder combined mean-p (orange); bands are 95% cluster-bootstrap intervals over cosmologies (the window is re-evaluated in each draw). Triangles mark the 50%-power crossings of the KL on NLA.

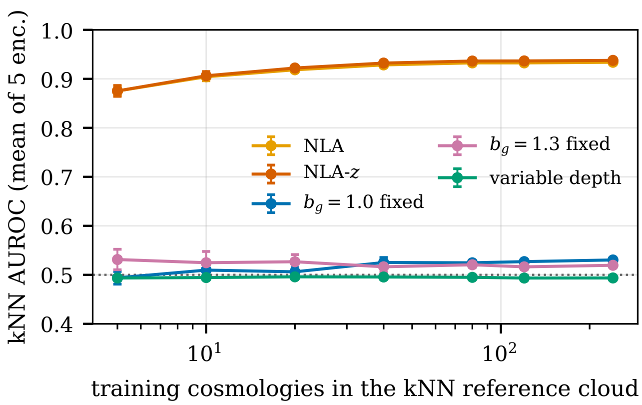

Figure 3: The kNN detector needs only a small reference cloud. Single-encoder kNN AUROC (mean over five encoders) after refitting whitening and tree on random subsets of the training cosmologies (all augmentations kept); error bars are the sd over eight subsets. Full cloud: 240 cosmologies, 19167 summaries.

consistent with the 5% false-alarm rate. The change of *model form*, from a mass- and red-fraction-dependent alignment to a uniform one, is therefore not detectable at matched effective amplitude; only the amplitude offset is.

Figure 2b turns this into a threshold. Walking outward from the training median, half of all NLA mocks are detected at $A_{\rm IA}^{\rm total} = -0.35$ $[-0.55, -0.30]$ below and 1.50 [1.15, 1.65] above for kNN, and at -0.55 $[-0.65, -0.45]$ and 1.90 [1.70, 2.25] for the KL: $|\Delta A_{\rm IA}^{\rm total}| = 0.78$ (kNN) and 0.98 (KL) towards zero, 1.07 and 1.47 towards stronger alignment. NLA-z gives the same picture (50% power at -0.65 and 1.25 for kNN, -0.90 and 1.80 for KL). Reliable detection, 80% power, needs $A_{\rm IA}^{\rm total} \leq -0.70$ or ≥ 2.35 (kNN, NLA), i.e. $|\Delta A_{\rm IA}^{\rm total}| \gtrsim 1.13$ downward and 1.92 upward; the KL needs 1.33 and 3.22. The asymmetry arises because the training amplitude is strictly positive (minimum 0.27): an alignment of the wrong sign has no counterpart in the training suite, whereas a stronger positive one differs only in degree. Mean-p is markedly weaker on the high side (never 50% power for NLA within $A_{\rm IA}^{\rm total} \leq 6$), the conditional score's loss of IA conditioning noted above. In short, *one unit of* $A_{\rm IA}^{\rm total}$ *is the resolution of these detectors at 50% power; about* 1.2 *units towards zero and* 2 *(kNN) to* 3 *(KL) upward give reliable detection.* Half a unit is mostly missed: at $|\Delta A_{\rm IA}^{\rm total}| = 0.5$ the kNN power is 0.22 [0.10, 0.37] towards zero and 0.31 [0.19, 0.44] upward (KL: 0.09 and 0.18).

3.3 What is not detectable, and why it matters

On the scale of Fig. 2b, the variable-depth and b_g suites sit at the false-alarm floor: their powers (0.06, 0.05, 0.04 for KL; 0.05, 0.05, 0.08 for kNN) are indistinguishable from the in-band IA mocks, i.e. below the smallest IA offset the curve resolves, and their AUROC intervals include 0.5 (Table 1). The truth-requiring TARP error agrees once compared with a floor drawn at each suite's own size: at 40 cosmologies × 2 mocks the in-distribution floor is $\mathcal{C} = 0.0560 \pm 0.0105$ (95th percentile 0.0734), against which $b_g = 1.0$ (0.0618) is +0.6 sd and $b_g = 1.3$ (0.0511) −0.5 sd; variable depth is +0.4 sd from its own floor. None is a detection, while the IA suites are hundreds of sd above theirs ($\mathcal{C} = 1.053$, 0.921).

For b_g this is largely the correct answer. Against the training prior, $b_g = 1.3$ lies at $z = +1.6, +1.5, +1.4, +0.6, -0.8, -1.8$ in bins 1–6: inside the support of every bin, with a joint $\chi^2 = 11.0$ for 6 degrees of freedom ($p = 0.089$), an unremarkable draw from the prior. $b_g = 1.0$ is out of support in the two highest-redshift bins ($z = -3.9, -4.9$; $\chi^2 = 46.8$, $p = 2.1 \times 10^{-8}$), a genuinely out-of-distribution configuration, and it is accordingly the one hard suite with a marginal conditional signal. That even this configuration is invisible to the unconditional kNN and to the KL says that a coherent, mild shift of a nuisance field leaves the 8-dimensional summary inside the training cloud.

What matters is what these slices do to the posterior. Because the two b_g suites and the in-distribution suite share 80 identical mocks (same cosmologies, rotations and noise realisations), the posterior bias can be compared per mock. In units of the posterior width, the mean of $(\theta_{\rm true} - \mu_{\rm post})/\sigma_{\rm post}$ over these mocks is +0.24 [+0.11, +0.37] in $\Omega_{\rm m}$ and −0.07 in σ_8 for the training physics, but −0.79 [−0.91, −0.67] and +0.94 for $b_g = 1.0$, and +0.77 [+0.67, +0.88] and −0.88 for $b_g = 1.3$ (means over the five encoders, which agree to ≤ 0.1). The bias is antisymmetric in b_g and equal in size for the in-support and the out-of-support slice: the posterior mean tracks the true galaxy bias linearly, by $\Delta\Omega_\mathrm{m} = -0.0489 \pm 0.0032$, $\Delta\sigma_8 = +0.0757 \pm 0.0069$ and $\Delta w_0 = +0.0408 \pm 0.0074$ between the two suites (-1.49, $+1.84$ and $+0.28$ posterior widths; in-distribution $\sigma_\mathrm{post} = 0.0329$, 0.0411, 0.145). Two facts constrain the reading. The TARP calibration of the in-support slice is at the in-distribution floor, so the implicit marginalisation over b_g has not failed in the sense of mis-stated posterior widths. And a b_g-marginalised posterior is calibrated only on average over the b_g prior: at any *fixed* b_g its mean is conditionally biased by however much the summaries are degenerate with b_g, which here is a shift of $0.8\,\sigma_\mathrm{post}$ in Ω_m and σ_8 at b_g values that the prior itself considers ordinary. The out-of-support slice adds no measurable extra bias beyond this linear trend. The trend does not, however, fully account for the training physics: the line through the two slices predicts $+0.35$ in Ω_m and -0.39 in σ_8 at the prior-mean bias $b_g = 1.22$, against $+0.24$ [$+0.11$, $+0.37$] and -0.07 [-0.17, $+0.04$] observed. Part of the slices' bias is therefore specific to what both share and the prior rarely produces: one value of b_g in all six bins, a coherent tilt against the prior's rise of b_g with redshift. In that sense the slices are mildly out-of-distribution in a way no summary-space score registers. For comparison, the DES Y3 map-level analysis [3] bounds its source-clustering sensitivity at $|\Delta\Omega_\mathrm{m}| < 0.007$ with a narrower posterior (0.024); ours is 7.0× larger with a wider one. No detector can catch this, because nothing about such an observation is unusual: it is the price of a summary that carries source-clustering information without a parameter to absorb it. Inferring $b_g^{(1..6)}$ explicitly would turn the conditional bias into a reported b_g–Ω_m degeneracy and let the conditional OOD score see a pinned bias as a parameter shift.

4 Conclusions

1. **OOD beats KL, at a fifth of the cost.** One encoder's kNN distance in its own summary space separates IA-misspecified mocks with AUROC 0.93 and 80% power at 5% false alarms; the five-encoder posterior-disagreement KL reaches 0.89 and 0.74. Both flag the same mocks and neither adds to the other; the kNN reference needs only a few hundred training summaries. The KL remains useful when no summary cloud is retained, but needs ≥ 2 encoders and is the weaker statistic.

2. **The detection threshold is about one unit of $A_\mathrm{IA}^\mathrm{total}$:** 50% detection at $|\Delta A_\mathrm{IA}^\mathrm{total}| \approx 0.8$–1.0 from the training median, 80% at ≈ 1.1–1.3 towards zero and ≈ 2 (kNN) to 3 (KL) upward; a change of IA model form at matched amplitude is undetectable.

3. **Variable depth and the galaxy-bias slices are not detectable**, and for $b_g = 1.3$ it should not be: it is inside the training prior; $b_g = 1.0$ is clearly outside in the joint prior ($p = 2.1 \times 10^{-8}$), though only through bins 5–6, and still leaves no trace. The actionable result is that the posterior mean tracks the true b_g ($\Delta\Omega_\mathrm{m} = -0.0489$ over the two slices, 7.0× the DES Y3 bound, $\pm 0.8\,\sigma_\mathrm{post}$ at fixed b_g) while its calibration stays at the floor: partly the conditional bias of a b_g-marginalised posterior at fixed b_g, partly the response to a coherent across-bin b_g configuration the prior rarely produces. Neither is flagged by any OOD statistic, and both must be addressed on the inference side (an explicit or wider b_g model), not with a better detector.

4. **Limitations.** Results hold for this encoder family at 300 training cosmologies; the dependence on *training*-set size (as opposed to reference-cloud size) was not measured. The b_g suites have 40 cosmologies, so AUROC differences below $\sim$0.07 are unresolved there. On real data the kNN score is evaluated as is; the conditional score needs posterior draws of θ.

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